



TURNING



Example 1

A car main gear shaft is made using a 18NiCrMo4 steel. Some operations are done by using a lathe, using a between centers configuration.

The first operation brings the diameter of the part from 48 mm to 40 mm. The stock is 180 mm long. This operation is divided in two steps: roughing and finishing.

Operation data: $k_{c0,4} = 2.800$ MPa, coefficient $x = 0,29$, entering angle of the main cutting edge $\kappa_r = 35^\circ$, entering angle of the secondary cutting edge $\kappa_r' = 55^\circ$

- **Roughing.** The tool used has a tip radius $r_\epsilon = 1,2$ mm:

Feed	Depth of cut	Cutting speed
$f = 0,5$ mm/rev	$a_p = 3$ mm	$v_c = 170$ m/min

- **Finishing.** The tool used has a tip radius $r_\epsilon = 0,4$ mm :

Feed	Depth of cut	Cutting speed
$f = 0,2$ mm/rev	$a_p = 1$ mm	$v_c = 325$ m/min



Example 1

2

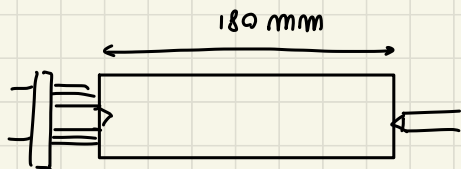
The lathe has a nominal power of 25 kW and a 90% efficiency; the maximum number of rpm is equal to 5000 rpm.

It is required to:

- a) Check if the machine specifications are compatible with the execution of the required machining operations.
- b) Determine the theoretical roughness after the finishing step.
- c) Determine the machining time for the finishing step, considering an overtravel of 2 mm before and after the operation.

extra path IN-OUT

EX1 manufacturing of a shaft 1m between center condition



$$K_{c0.4} = 2800 \text{ MPa}$$

$$\alpha = 0.29$$

(KRONENBERG) } for K_c
CUTTING PRESSURE

$$\begin{cases} K_r = 35^\circ \\ K_r' = 55^\circ \end{cases}$$

Machinability data

$$P_e = 25 \text{ kW}$$

$$\eta = 90\%$$

$$m_{\text{MAX}} = 5000 \text{ rpm}$$

$$D_o = 48 \text{ mm}$$

$$D_f = 40 \text{ mm after finishing}$$

ROUGHING

$$a_p = 3 \text{ mm}$$

$$f = 0.5 \text{ mm/REV}$$

$$V_c = 170 \text{ m/min}$$

(you remove BIG amount of material with limited time)

$$r_E = 1.2 \text{ mm (ROBUST)}$$

$$e_{\text{IN}} = e_{\text{OUT}} = 2 \text{ mm}$$

extra path for machining

FINISHING

$$a_p = 1 \text{ mm}$$

(from 48 we remove 3 of radial depth of cut, after ROUGHING:

$$D_R = 48 - 2 \cdot 3 = 42 \text{ mm}$$

and to reach 40 mm we need to remove $42 - 2 \cdot 2 = 40$
IT COULD BE IMPLIC IT!

$$f = 0.2 \text{ mm/REV}$$

$$V_c = 325 \text{ m/min}$$

(small amount of material accurately removed faster)

V_c depends on CUTTED Material

$$r_E = 0.4 \text{ mm}$$

a check compatibility with specification

$$\begin{cases} P_c \leq 0.9 \cdot 25 = 22.5 \text{ kW} & \text{LIMIT ON POWER} \\ m \leq m_{\text{MAX}} \end{cases}$$

ROUGHING (R) $F_c = K_c \cdot A_D$

$$A_D = f \cdot a_p = 0.5 \cdot 3 = 1.5 \text{ mm}^2$$

$$K_c = (\text{respect the specific } K_{c0.4}) = K_{c0.4} \cdot \left(\frac{0.4}{f \sin(K_r)} \right)^\alpha = 2800 \left(\frac{0.4}{0.5 \sin(35)} \right)^{0.29} = 3083.63 \text{ MPa}$$

When $K_r = 90^\circ$, $a_p = 0.4$

$$F_c = 3083.63 \cdot 1.5 = 4625.45 \text{ N}$$

$$P_c = F_c \cdot V_c = \frac{4625.45 \cdot [170]}{60 \cdot 1000} \text{ [m/min]} \rightarrow \text{to obtain kW we divide by 1000}$$

$$= 13.1 \text{ kW} < 22.5 \text{ kW} \quad \underline{\text{OK}} \quad \text{POWER Feasibility!}$$

$$\text{check for the RPM: } n = \frac{1000 \cdot V_c}{\pi D_R} = \frac{1000 \cdot 170}{\pi \cdot 42} = 1288 \text{ rpm} < 5000 \quad \underline{\text{OK}} \quad m_{\text{MAX}}$$

FINISHING in the same way

$$A_b = 0,2 \cdot 1 = 0,2 \text{ mm}^2 \text{ (less than Roughing Area)}$$

$$K_c = 2800 \left(\frac{0,4}{0,2 \cdot \sin(35)} \right)^{0,29} = 4022,21 \text{ MPa}$$

$$\Rightarrow F_c = 4022,21 \cdot 0,2 = 804,44 \text{ N}$$

smaller than the Roughing one, so less mass

$$P_c = \frac{804,44 \cdot 325}{60 \cdot 1000} = 4,36 \text{ kW} < 22,5 \text{ kW} \quad \text{OK}$$

$$n = \frac{1000 \cdot 325}{\pi \cdot 40} = 2586 \text{ rpm} < 5000 \text{ } n_{\text{max}} \quad \text{OK}$$

Feasible OVERALL! || both R+F feasible!

(b) theoretical roughness after finishing

↓ we come about this after finishing! final STEP

find expected average Roughness

having $r_z = 0,4 \text{ mm}$ TOOL RADIUS "r"

apply SWARTZ formula?

$$R_a = \frac{f^2}{32r} \cdot 10^3 [\mu\text{m}] = \frac{(0,2)^2}{32 \cdot 0,4} 1000 =$$

$$= 3,125 \mu\text{m}$$

BUT, to use this

formula, VERIFY if

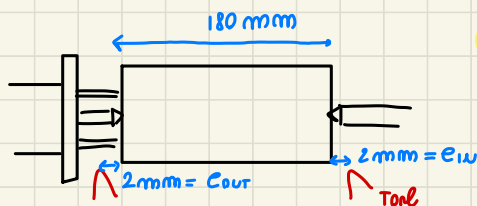
basic hypothesis satisfied (to have circular profile)

$$\text{Hyp: } \left\{ \begin{array}{l} \| f/2 \leq r \sin(\kappa_r) \| \rightarrow \kappa_r < \kappa'_r \text{ it is relevant constrain } \sin(\kappa_r) < \sin(\kappa'_r) \\ f/2 \leq r \sin(\kappa'_r) \end{array} \right. \downarrow f/2 \leq r \sin(\kappa_r)$$

$$0,2/2 = 0,1 \leq 0,4 \sin(35^\circ) = 0,229 \quad \text{OK}$$

(c) machining time for finishing operation

{ we can apply the R_a eq.
we have a profile as a SET of ARCS }



$$t_m = \frac{2 + 180 + 2}{517,2} = 0,3558 \text{ min} = 21,4 \text{ s}$$

$$t_m = \frac{e_{\text{TOT}}}{V_f}$$

$$\left\{ \begin{array}{l} V_f = f \cdot n = 0,2 \cdot 2586 = \\ = 517,2 \text{ mm/min} \end{array} \right.$$

$V_f < V_c$ so we can neglect P_f of feed motion power!

$$\Leftarrow \left\{ \begin{array}{l} V_f = 0,517 \text{ m/min} \\ V_c = 325 \text{ m/min} \end{array} \right. \quad \text{COMPARISON...}$$



Example 2

A cylindrical component is machined using a lathe.

Process parameters:

- Tip radius $r = 0,4 \text{ mm}$
- Roughness $R_a = 3,2 \text{ }\mu\text{m}$
- Depth of cut $a_p = 2 \text{ mm}$
- Specific cutting force $k_{c 0,4} = 2100 \text{ MPa}$
- Constant $x = 0,29$
- Entering angles: $\kappa_r = \kappa_r' = 45^\circ$
- Taylor's Law parameters: $C = 250, n = 0,122$

Machine parameters:

- Nominal power: 6 kW
- Efficiency: 80%
- T_t : 120 s

- a) Determine the cutting speed that maximizes the throughput.
- b) Determine the maximum feed, considering all constraints.

PROCESS

TAYLOR'S LAW

$$a_p = 2 \text{ mm}$$

$$r = 0.4 \text{ mm}$$

$$R_a = 3,2 \mu\text{m}$$

$$K_r = K_r' = 45^\circ$$

KRONENBERG

$$\begin{cases} K_{co.4} = 2100 \text{ MPa} \\ \chi = 0.29 \end{cases}$$

MACHINE

$$P_e = 6 \text{ kW}$$

$$\eta = 80\%$$

$$T_t = 120 \text{ s}$$

tool changing time

V_c ?

f ?

CONSTRAINTS
(Goal to guarantee)

- (a) V_c to have MAX THROUGHPUT
- (b) f_{MAX} ? considering all constraints

(a) $\max\{\text{throughput}\}$

(it is the amount of part produced per unit of time) so we MAXIMIZE throughput BY minimizing production time

$\min\{T_p\}$

↓

$$T_{OPT} = T_t \left(\frac{1}{m} - 1 \right)$$

this result of optimization is in [min]

so we should buy all in [min]

$$T_t = 120/60 = 2 \text{ min}$$

$$= 2 \left(\frac{1}{0.122} - 1 \right) = 14,39 \text{ min (usually } \approx 15 \text{ min) optimal tool life}$$

so KNOWING T_{OPT} production time...

$$V_{COPT} = \frac{C}{T_{OPT}^m} = \frac{250}{(14,39)^{0.122}} = 180,57 \frac{\text{m}}{\text{min}}$$

cutting speed that

maximize the throughput (OPTIMIZATION Result)

(b) maximum feed possible

respect all constraints

$$\begin{cases} P \leq P_e \eta \\ R \leq R_a \end{cases}$$

(from DATA we KNOW ROUGHNESS INFO)

ROUGHNESS: limit

$$R_a = \frac{f^2}{32 r} 1000 \Rightarrow$$

$$f_{MAX} = \sqrt{\frac{32 \cdot r \cdot R_a}{1000}} = \sqrt{\frac{32 \cdot (0,4) \cdot 3,2}{1000}} = 0,2 \text{ mm/REV}$$

↓
this is valid only

IF the hyp $f/2 < \min\{r_{sm}(K_r), r_{sm}(K_r')\}$

to verify after verification...

$$\begin{cases} \frac{f_{MAX}}{2} \leq r_{sm}(K_r) \\ \frac{f_{MAX}}{2} \leq r_{sm}(K_r') \end{cases} \Rightarrow$$

$$\frac{0,2}{2} \leq 0,4 \sin(45)$$

$$0,1 \leq 0,28 \text{ OK}$$

hypot valid for f formula

While respect POWER limit.

$$P_c \leq \eta P_e \Rightarrow$$

$$P_c = \frac{F_c \cdot V_c}{60 \cdot 1000}$$

Power limit

$$F_c = K_{c0.4} \left(\frac{0.4}{\sin(\kappa_r)} \right)^x a_p f^{1-x}$$

$$\frac{F_c v_c}{60 \cdot 1000} \leq \eta P_e$$

$$\Rightarrow K_{c0.4} \left(\frac{0.4}{\sin(\kappa_r)} \right)^x a_p f^{1-x} \frac{v_c}{60000} \leq \eta P_e$$

$$f \leq \left(\frac{\eta P_e 60000 (\sin(\kappa_r))^x}{v_c K_{c0.4} (0.4)^x a_p} \right)^{\frac{1}{1-x}} = \left(\frac{0.8 \cdot 6 \cdot 60000}{180.57 \cdot 2100 \left(\frac{0.4}{\sin(45)} \right)^{0.29} 2} \right)^{\frac{1}{1-0.29}} =$$

$v_{c0.07}$ found before

$$= 0.323 \text{ mm/REV}$$

$$\downarrow \begin{cases} f \leq 0.323 \text{ mm/REV} & \text{Power} \\ f \leq 0.2 \text{ mm/REV} & \text{Roughing} \end{cases}$$

$\uparrow$ LIMIT $\Rightarrow$

$$f_{MAX} = 0.2 \text{ mm/REV}$$



Example 3

until now external cylindrical turning

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A component is machined using a lathe from a steel bar, using a between centers configuration. In the following table, some machining parameters are reported.

Product data	Process Parameters	Process times	Costs
$L = 200 \text{ mm}$	$f = 0,5 \text{ mm/rev}$	$T_t = 60 \text{ s}$	$C_0 = 100 \text{ €/h}$
$D_0 = 40 \text{ mm}$	$\kappa = 45^\circ$	$T_h = 30 \text{ s}$	$C_t = 5 \text{ €}$
$D_f = 36 \text{ mm}$	$n = 0,122$		
	$C = 230$		
	$k_{c0,4} = 2600 \text{ MPa}$		
	$x = 0,29$		

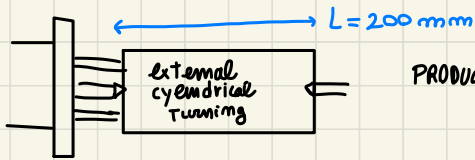


Example 3

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Determine:

- a) Cutting power with a cutting speed of $v_c = 130$ m/min;
- b) Cutting speed such that it is possible to make exactly 10 parts;
- c) Check production costs of points a) and b);
- d) Cutting speed that minimizes the production cost, and, therefore, the optimal cost.



PRODUCT

$$\begin{cases} D_o = 40 \text{ mm} \\ D_f = 36 \text{ mm} \\ f = 0.5 \text{ mm/REV} \\ Kr = 45^\circ \end{cases}$$

(SINGLE cutting edge)

BRING to C mm/min

$$C_o = 1.67 \text{ €/min}$$

↑ (± 80)

KRONENBERG

TAYLOR

$$\begin{cases} K_{10.4} = 26000 \text{ MPa} \\ \chi = 0.29 \end{cases}$$

$$\begin{cases} C = 230 \\ m = 0.122 \end{cases}$$

times

$$\begin{cases} T_t = 60 \text{ s} \\ T_h = 30 \text{ s} \end{cases}$$

costs

$$\begin{cases} C_o = 100 \text{ €/h} \\ C_t = 5 \text{ €} \end{cases}$$

(tool cost)

- a) P_c ? of $V_c = 130 \text{ m/min}$
 b) V_c so that we can product exactly $m_p = 10$ parts?
 c) check C_{pa} , C_{pb} ?
 d) V_c that minimize C_p ?
- OPTIMIZATION questions

a) $a_p = \frac{D_o - D_f}{2} = \frac{40 - 36}{2} = 2 \text{ mm}$

$$K_c = K_{10.4} \left(\frac{0.4}{f \sin(Kr)} \right)^{\chi} = 2600 \left(\frac{0.4}{0.5 \sin(45^\circ)} \right)^{0.29} = 2694.75 \text{ MPa}$$

$$A_D = 0.5 \cdot 2 = f \cdot a_p = 1 \text{ mm}^2$$

$$F_c = K_c \cdot A_D = 2694.75 \cdot 1 = 2694.75 \text{ N}$$

$$\Rightarrow P_c = \frac{F_c \cdot V_c}{60 \cdot 1000} = \frac{2694.75 \cdot 130}{60 \cdot 1000} = 5.84 \text{ kW}$$

- b) V_c so that tool life enables you to product exactly 10 PARTS

→ We require to have enough tool

LIFE (T) to make 10 parts ⇒

$$T = 10 \cdot t_c$$

(considering only CONTACT time)

1 PART → t_c is the time when WEAR happens!

from here we use
KNOWN relationship TO
express all respect V_c

$$t_c = \frac{L_c}{f \cdot m}$$

cutting length

$$m = \frac{1000 \cdot V_c}{\pi D_f}$$

⇒

$$T = \left(\frac{L_c}{f \cdot \frac{1000 V_c}{\pi D_f}} \right) m_p$$

t_c

TIP POINT
WHERE IT CUT!

Where we know the expression of T
by TAYLOR'S LAW related to V_c ⇒

$$V_c \cdot T^m = C \rightarrow V_c \left(\frac{m_p L_c \cdot \pi D_f}{f \cdot 1000 V_c} \right)^m = C$$

⇒ so from here we have

$$V_c^{\frac{1}{1-m}} = \left[C \left(\frac{1000 \cdot f}{m_p L_c \cdot \pi \cdot D_f} \right)^m \right]^{\frac{1}{1-m}} = \left[230 \left(\frac{1000 \cdot 0.5}{10 \cdot 200 \cdot \pi \cdot 36} \right)^{0.122} \right]^{\frac{1}{1-0.122}}$$

$$= 209.4 \text{ m/min}$$

④ Compute cost C_p in the two discussed situation (a) and (b)

from the equation $C_p = C_o \cdot T_h + C_o T_c \left[1 + \frac{1}{T} \left(T_c + \frac{C_t}{C_o} \right) \right]$

CASE (a): $V_c = 130 \text{ m/min}$

$$T = \left(\frac{230}{130} \right)^{\frac{1}{0.122}} = 107.4 \text{ min (long tool life)}$$

$$n = \frac{1000 \cdot 130}{\pi \cdot 36} = 1149 \text{ rpm} \quad t_c = \frac{200}{0.5 \cdot 1149} = 0.35 \text{ min}$$

$$C_{pa} = 1.67 \cdot 0.5 + 1.67 \cdot 0.35 \left[1 + \frac{1}{107.4} \left(1 + \frac{5}{1.67} \right) \right] = 1.44 \text{ €}$$

CASE (b): $V_c = 209.4 \text{ m/min}$

$$n = \frac{1000 \cdot 209.4}{\pi \cdot 36} = 1852 \text{ rpm} \quad t_c = \frac{200}{0.5 \cdot 1852} = 0.216 \text{ min}$$

$$T = \left(\frac{230}{209.4} \right)^{\frac{1}{0.122}} = 2.16 \text{ min} = 10 t_c \text{ as wanted on (b)}$$

↓

same equation of

difference respect (a)

$$C_{pb} = 1.67 \cdot 0.5 + 1.67 \cdot (0.216) \left[1 + \frac{1}{(2.16)} \left(1 + \frac{5}{1.67} \right) \right] = 2.28 \text{ €}$$

d) v_c such C_p is min.

$$T_{\text{opt}}(C_p) = \left(T_t + \frac{C_t}{C_o} \right) \left(\frac{1}{n} - 1 \right) = \left(1 + \frac{5}{1,67} \right) \left(\frac{1}{0,122} - 1 \right) = 28,74 \text{ min}$$

TO OPTIMIZE
THE COST

$$v_{\text{copt}}(C_p) = \frac{C}{T_{\text{opt}}(C_p)^m} = \frac{230}{(28,74)^{0,122}} = 152,7 \frac{\text{m}}{\text{min}}$$

C_p ?

$$n = \frac{1000 \cdot 152,7}{\pi \cdot 36} = 1350 \text{ rpm}$$

$$t_c = \frac{200}{0,5 \cdot 1350} = 0,3 \text{ min}$$

$$C_p = 1,67 \cdot 0,5 + 1,67 \cdot 0,3 \left(1 + \frac{1}{28,74} \left(1 + \frac{5}{1,67} \right) \right) = 1,41 \text{ €}$$

for a single
part for JUST one operation...

When mass production, also small C_p
differences are relevant to spend less $\rightarrow$ ECONOMICAL Benefits!



Example 4

simple object

A shop floor produces **steel cylindrical shafts**. An **external cylindrical turning** is done using a **between centers** configuration. It brings the diameter of the part from 12 mm to 10 mm. The part is 70 mm long.

Machining parameters:

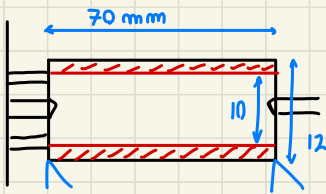
- **Roughing:**
 - Depth of cut $a_p = 0,8 \text{ mm}$;
 - Feed $f = 0,2 \text{ mm/rev}$;
 - Cutting speed $v_c = 150 \text{ m/min}$;
 - Extra times = 2 s;
 - Tool life = 10 min.
 - Tool change time = 3 min.
 - Load/unload time = 10 s.
- **Finishing:**
 - Total time = 10 s;



Example 4

Determine:

- a) Contact time for the roughing operation;
- b) Roughing tool life, using a cutting speed of 130 m/min , knowing that if the cutting speed is reduced at 125 m/min , the corresponding tool duration is equal to 25 min ;
- c) Number of shafts produced in one shift of 8 hours, using a cutting speed of 150 m/min (assume infinite tool life for finishing and cutoff operations). The setup time is equal to 15 min , at the beginning of each shift.



ROUGHING: $a_p = 0,8 \text{ mm}$ $f = 0,2 \text{ mm/REV}$

$$V_c = 150 \text{ m/min}$$

$$\text{extra time} = 2 \text{ s}$$

$$T = 10 \text{ min} \quad T_t = 3 \text{ min}$$

$$T_L = 10 \text{ s}$$

(a) t_c ? on Roughing

(b) T ? if $V_c = 130 \text{ m/min}$
consider that when $V_c = 125 \text{ m/min} \rightarrow T = 25 \text{ min}$

(c) m_p ? working & h with $V_c = 150 \text{ m/min}$

(a) $l_c = L_{\text{tot}} = 70 \text{ mm}$ (between center)

$$t_c = \frac{l_c}{n \cdot f} = \frac{70}{(\overset{\text{on Roughing}}{n}) \cdot 0,2} = \frac{70}{4591 \cdot 0,2} = \frac{D_R}{4591 \cdot 0,2} = \frac{D - 2a_p}{4591 \cdot 0,2} = \frac{12 - 2 \cdot 0,8}{4591 \cdot 0,2} = 10,4 \text{ mm}$$

AFTER ROUGHING

$$n = \frac{1000 \cdot V_c}{\pi \cdot D_R} = \frac{1000 \cdot 150}{\pi \cdot 10,4} = 4591 \text{ rpm}$$

$$t_c = \frac{70}{4591 \cdot 0,2} \cdot 60 = 4,6 \text{ s}$$

(b) to compute TOOL LIFE when $V_c = 130 \text{ m/min}$...

WE need the TAYLOR's parameters!

to evaluate

taylor law data

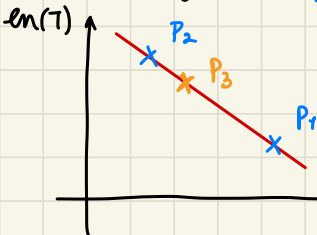
we use the two KNOWN points

$$\left\{ \begin{array}{l} (T = 10 \text{ min}, V_c = 150 \text{ m/min}) \text{ POINT 1 (P}_1\text{)} \\ (T = 25 \text{ min}, V_c = 125 \text{ m/min}) \text{ POINT 2 (P}_2\text{)} \end{array} \right\}$$

find what we need...

from the KNOWN POINTS...

consider



$$V_c T^m = C \quad \text{Taylor's law}$$

$$\Rightarrow \ln(V_c) + m \ln(T) = \ln(C)$$

$$\left\{ \begin{array}{l} 1) \ln(V_{c1}) + m \ln(T_1) = \ln(C) \\ 2) \ln(V_{c2}) + m \ln(T_2) = \ln(C) \end{array} \right.$$

$$\downarrow$$

$$\ln(V_{c1}) + m \ln(T_1) = \ln(V_{c2}) + m \ln(T_2)$$

$$m (\ln(T_1) - \ln(T_2)) = \ln(V_{c2}) - \ln(V_{c1})$$

$$m = \frac{\ln(V_{c2}/V_{c1})}{\ln(T_1/T_2)} = \frac{\ln(125/150)}{\ln(10/25)} = 0,199$$

using directly

$$C = V_c T^m = V_{c1} T_1^m = (150) \cdot 10^{0,199} = 237,19$$

once found m, C TAYLOR'S constant

$$\downarrow$$

$$T_3 = (C/V_{c3})^{\frac{1}{m}} = (237,19/130)^{\frac{1}{0,199}} = 20,53 \text{ min}$$

on our third
WORKING condition

We find $P_3 \in [P_1, P_2]$ inside our Range!

This is important $\rightarrow$ looking T between

18 T outside our Range (10,25) it will be wrong

(MP)

 PART produced in a shift of 8 hours = T_{tot} considering $V_c = 150 \text{ m/min}$
and all the other lost time

$$T_p = T_h + T_c \left(1 + \frac{T_t}{T}\right)$$

FIXED CONTACT

$$\Downarrow$$

$$T_p = 22 + 4,6 \left(1 + \frac{180}{600}\right) = 27,98 \text{ s} \approx 28 \text{ s}$$

$$\left\{ \begin{array}{l} T_c (V_c = 150 \text{ m/min}) = 4,6 \text{ s} \\ \text{ (as computed on (a))} \\ T_h = 2 \text{ s} + 10 \text{ s} + 10 \text{ s} = 22 \text{ s} \\ \text{EXTRA LOAD/ UNLOAD FINISHING AS CONSTANT} \\ T_t = 3 \text{ min} = 180 \text{ s} \\ T = 10 \text{ min} = 600 \text{ s} \end{array} \right.$$

so evaluated T_p time $\forall$ PART...

$$M_p = \left\lfloor \frac{T_{available}}{T_p} \right\rfloor = \left\lfloor \frac{8 \cdot 3600 - 15 \cdot 60}{28} \right\rfloor =$$

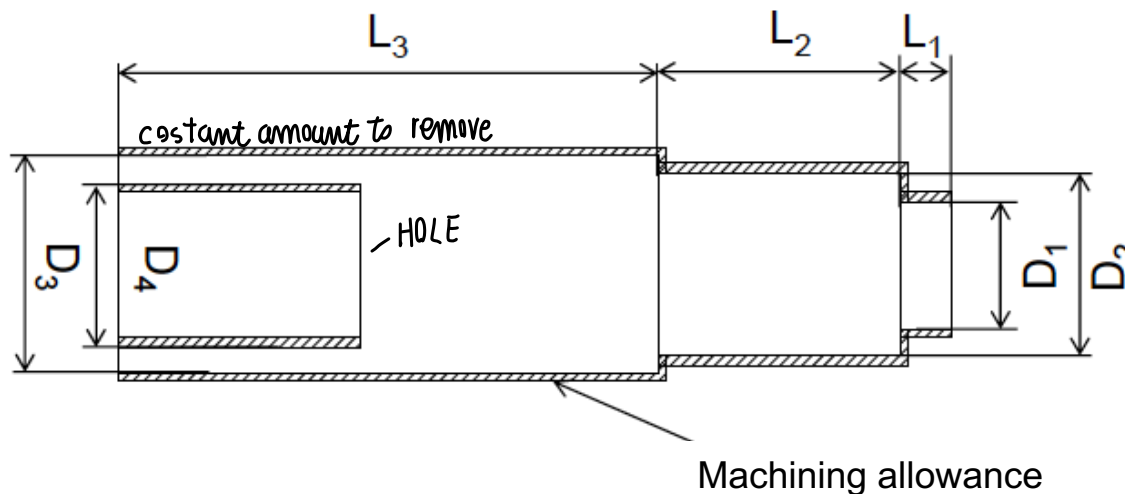
NUMBER of PRODUCED PART Floor operator preparation theoretical Available time

$$= \lfloor 996,43 \rfloor = 996 \text{ PARTS}$$



Example 5

The figure reports the components to be machined starting from a steel stock. The dimensions are reported in the table. All sizes are referred to the finished part. Moreover, taking into account the interaction between cutting tool and material, the x and $k_{c0,4}$ coefficients are equal to 0,29 and 2100 MPa respectively.



Dimension	Value [mm]
D1	20
D2	25
D3	30
D4	24
L1	30
L2	70
L3	700

The machining allowance is equal to 1 mm and will be removed with a single finishing step. The average roughness R_a should be lower or equal than $1,5 \mu\text{m}$. The tool used for the finishing has a tip radius equal to 0,8 mm and an entering angle of the primary cutting edge equal to 95° and a secondary one equal to 45° .



Example 5

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The fixturing system consists in the usage of a self-centering three-jaws chuck, from the inside of the hole on the left. Using this fixturing, the finishing operation will be done on the external surfaces with a constant rpm, equal to 3000 rpm.

It is required to:



- a) Determine the contact time for the external finishing operation.
- b) Check the feasibility of the fixturing system for that operation.
- c) Considering two experimental cutting speed, namely v_{c1} and v_{c2} , and two experimental tool life values, namely T_1 and T_2 , determine the external finishing operation duration if the adopted cutting speed is set to the optimal value with respect to production time.

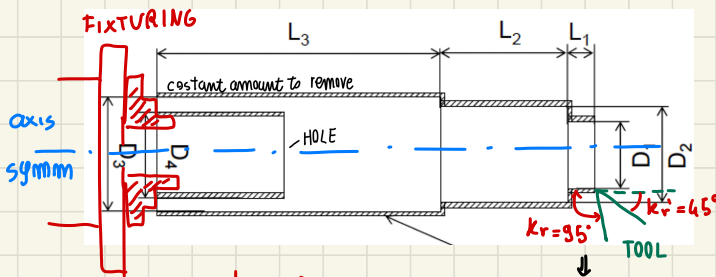


Example 5

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MORE DATA--

Friction coefficient	0,45
Number of jaws	3
Jaw-workpiece contact area	400 mm ²
Maximum contact pressure	100 MPa
V_{c1}	280 m/min
V_{c2}	310 m/min
T_1	19 min
T_2	12 min
T_t	60 s



HOLD using internal
surface: self-centering chuck

PERFORM FINISHING considering

$$M = 3000 \text{ rpm}$$

fixturing $Z = 3$ (Jaws numb.)

$$\mu = 0,45$$

$$A = 400 \text{ mm}^2 \text{ (contact area)}$$

$$P_{MAX} = 100 \text{ MPa}$$

(+ data on $V_{c1,2}; T_{1,2}$)

$D_1 = 20 \text{ mm}$ final D after
 $D_2 = 25 \text{ mm}$ machining
 $D_3 = 30 \text{ mm}$ remove material
 $D_4 = 24 \text{ mm}$ of surface in
 $L_3 = 700 \text{ mm}$ this config

$$L_2 = 70 \text{ mm}$$

$$L_1 = 30 \text{ mm}$$

$$K_{C0.4} = 21000 \text{ MPa} \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{KRONENBERG}$$

$$\chi = 0.29$$

$$\text{removing } a_p = 1 \text{ mm}$$

Requirements $R_a \leq 1,5 \text{ } \mu\text{m}$ limit!

$$\text{tool tip radius: } r = 0,8 \text{ mm}$$

⊙ contact time $t_c = \frac{L_c}{f \cdot m}$

$$L_c = \underset{\text{CONTACT LENGTH}}{30} + \underset{\text{axial}}{\frac{25-20}{2}} + \underset{\text{axial}}{70} + \underset{\text{Radial}}{\frac{30-25}{2}} + 700 = 805 \text{ mm}$$

We have to find f from Roughness R_a CONSTRAINTS!

$$R_a = \underset{\substack{\uparrow \\ \text{assuming} \\ \text{ARCS profile}}}{f^2} \frac{10^3}{32r} = \underset{\text{MAX}}{1,5 \text{ } \mu\text{m}} \text{ (limit)} \Rightarrow f_{MAX} = \sqrt{\frac{32 \cdot r \cdot 1,5}{1000}} =$$

$$\begin{cases} f/2 \leq r \sin(Kr) \\ f/2 \leq r \sin(K'r) \end{cases} \Rightarrow f \leq \min \left\{ 2r \sin(Kr), 2r \sin(K'r) \right\} = \min \left\{ 2 \cdot 0,8 \cdot \sin(45^\circ), 2 \cdot 0,8 \cdot \sin(95^\circ) \right\}$$

$$f \leq \min \{ 1,13; 1,13 \} = 1,13 \quad \underline{\text{OK}} \quad f \ll 1,13$$

$$\Rightarrow t_c = \frac{805}{0,196 \cdot \underline{3000}} = 82,2 \text{ s}$$

constant m means
that V_c changes along
the different geometry

(b)

 $T_R \geq T_C$ (Resistant torque $\geq$ cutting torque)! feasibility

$$T_C = F_c \cdot \frac{D/2}{1000} \leq T_R = Z \cdot \mu \cdot p_{\min} \cdot A \cdot \frac{D^*/2}{1000}$$

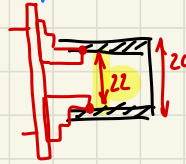
Diameter where the contact jaws-part happens

$$p \geq \frac{F_c \cdot D}{2 \mu A (D^*)} = \frac{506,8 \cdot 30}{3 \cdot 0,45 \cdot 400 \cdot 22} = 1,28 \text{ MPa} < p_{\max}$$

OK Feasible!

• own example: D^*

being $D_1 = 24 \text{ mm}$ BUT for $\Rightarrow D^* = D_1 - 2a_p = 24 - 2 = 22 \text{ mm}$
 machining allowance of 2mm

• to evaluate F_c

on external cyl cutting

$$F_c = K_c \cdot f \cdot a_p = 2585,5 \cdot 0,196 \cdot 1 = 506,8 \text{ N}$$

↓

$$K_c = 2100 \left(\frac{0,4}{0,196 \sin(95)} \right)^{0,29} = 2585,5 \text{ MPa}$$

 T_C ? Which torque we should consider? (which D use?)

We are cutting 3 different diameters, same force BUT radius respect axis is increasing $\rightarrow$ WORST CASE WHEN

 $D = D_3$ (max)

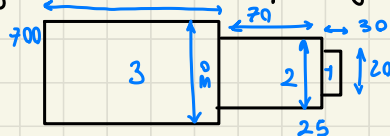
$$\left\{ \begin{aligned} T_{C\max} &= F_c \cdot \frac{D_3/2}{1000} \\ &= 506,8 \cdot \frac{30/2}{1000} \end{aligned} \right\}$$

(c) optimal $V_c(T_p)$ (respect production Time) knowing two points of Taylor's law

from DATA: (Taylor's law POINTS)

 $(V_{c1} = 280 \text{ m/min}, T_1 = 19 \text{ min}) P_1$ $(V_{c2} = 310 \text{ m/min}, T_2 = 12 \text{ min}) P_2$ $T_t = 60 \text{ s}$

from the DRAWING, speaking about optimal TOOL LIFE...



(SITUATION of WORK)

given 2 points we can find C, m of Taylor's Law

$$V_c T^m = C$$

$$\begin{cases} \ln(V_{c1}) + m \ln(T_1) = \ln(C) \\ \ln(V_{c2}) + m \ln(T_2) = \ln(C) \end{cases} \Rightarrow$$

$$m = \frac{\ln(V_{c2}/V_{c1})}{\ln(T_1/T_2)} = \frac{\ln(310/280)}{\ln(19/12)} = 0,22$$

Then using one point $C = V_{c1} T_1^m = 280 \cdot (19)^{0,22} = 535$
(approx same C if using V_{c2}, T_2 , a bit different)

FROM this information

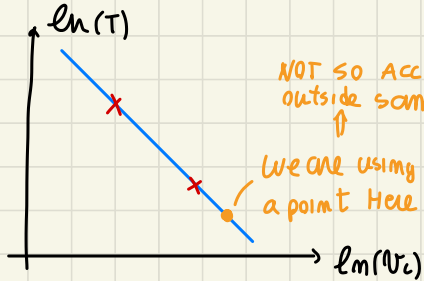
$\min\{T_p\} \Rightarrow T_{opt} = T_t \left(\frac{1}{m} - 1 \right) = 1 \left(\frac{1}{0,22} - 1 \right) = 3,55 \text{ min}$

$T_t = 69/60 = 1,15 \text{ min}$ ↑ use [min]!

(small optimal tool life!)

TAYLOR'S

Law $V_{c, opt} = C / T^m = 535 / (3,55)^{0,22} = 405 \text{ m/min}$



NOT SO ACCURATE!
outside sample

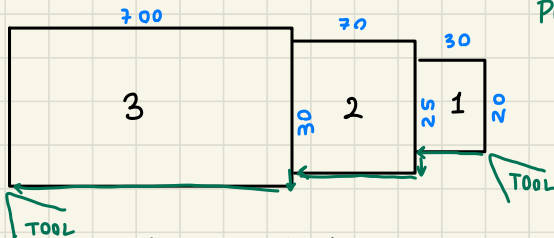
We are using
a point here!

→ to be sure I should do more experiment including optimal value
(out of Range of experimental points!)

HOW TO compute contact time

← With $V_c = \text{constant}$ in this situation?

Carefully, How to compute t_c , here we are moving the TOOL as:



PROFILING of the Workpiece
(as set of segments)

to maintain $V_c = V_{c, opt}$

WE have to change m
ROT SPEED on each cylinder!

↓

$$\begin{cases} m_1 = \frac{1000 \cdot 405}{\pi \cdot 20 \cdot 101} = 6446 \text{ rpm} \\ m_2 = \frac{1000 \cdot 405}{\pi \cdot 25 \cdot 102} = 5157 \text{ rpm} \end{cases}$$

$$m_3 = \frac{1000 \cdot 405}{\pi \cdot 30 \cdot 103} = 4297 \text{ rpm}$$

to keep v_c constant we vary (n) , constant n & cylindrical TURNING

m_1, m_2, m_3 const

(+ RADIAL STEPS) (*)

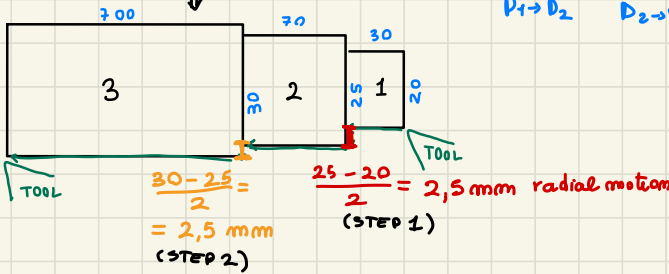
• JUST RESPECT external cylindrical turning (linear (axial) path)

$$t_c = \frac{L_1}{30} + \frac{L_2}{70} + \frac{L_3}{700} \quad (*)$$

f

$$+ \frac{2,5}{0,196 \cdot m_{12} (*)} + \frac{2,5}{0,196 \cdot m_{23} (**)}$$

radial motion to go from $D_1 \rightarrow D_2$
2 STEPS: $D_2 \rightarrow D_3$ ⇒ ISSUE here... to have $v_c = \text{const}$ during all cutting... m_{12}, m_{23} ?
Go from $D_1 \rightarrow D_2$ Go from $D_2 \rightarrow D_3$



(*) $m_{12} = \frac{m_1 + m_2}{2}$

(**) $m_{23} = \frac{m_2 + m_3}{2}$

We are going from D_1 to D_2 so $m_1 \rightarrow m_2$ (decreasing)

decrease m moving the tool radially ⇒ it is possible to compute $m_{12}(t)$ $\forall t$ along radial motion ⇒ we approx with mean value in between

(constant m while radially $v_c = \text{const}$ while $m \downarrow$)

done in lathe to have radial motion with v_c constant

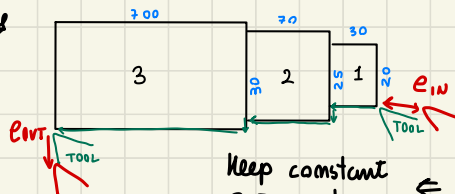
This is done by the CONTROL SYSTEM! controlling m during cutting

Carefully on the geometry!

IMPORTANT: If considering

t_m machining time

you have extra path in than e_{ov}



Keep constant ROT. speed

$\Leftarrow \begin{cases} e_w \text{ with } m_2! \\ e_{ov} \text{ with } m_3! \end{cases}$

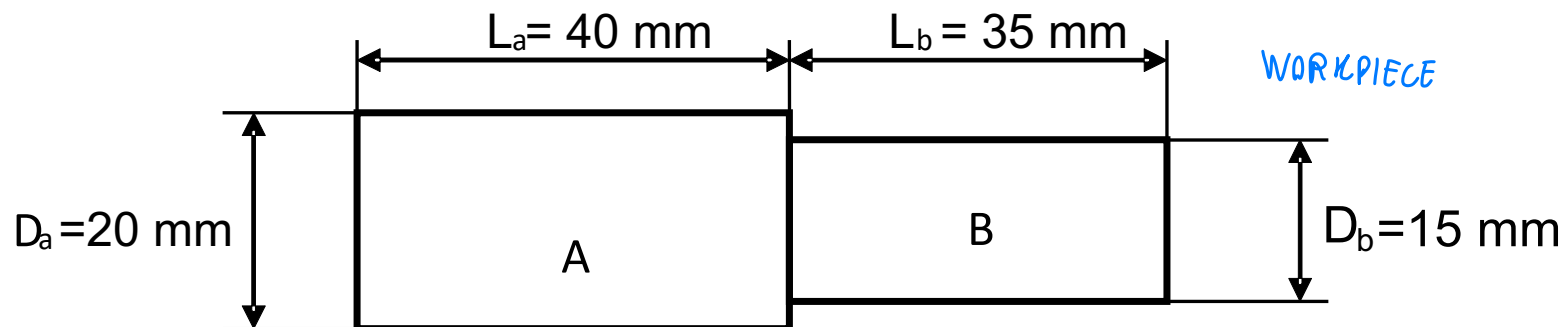
e_{ov}, e_{ov} extra path done in different conditions!

VAR4 CONTRIBUTE → due to diff condition



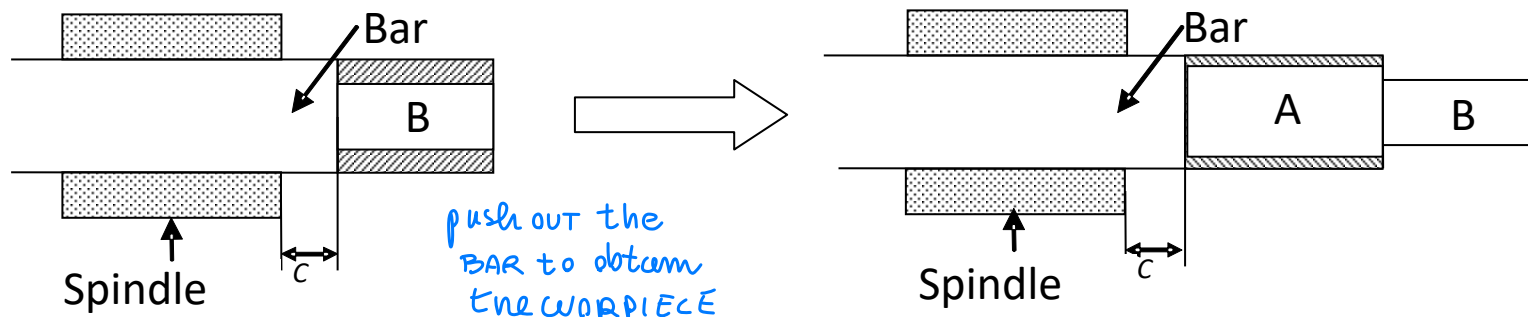
Example 6

The following component should be manufactured using a lathe, starting from a metal bar of diameter 25 mm. The bar is mounted using a lathe chuck.



The part can be machined by using four different strategies:

- **Strategy 1.** The length B is machined in one step. Then, the tool is disengaged from the part and the bar is repositioned as shown on the right side. The length A is machined in one step. The feed value is equal to 0,2 mm/rev, while the repositioning time is equal to 2 s.

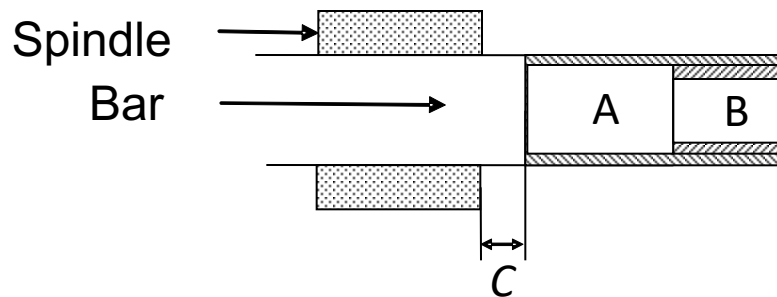




Example 6

12

- **Strategy 2.** Same as Strategy 1, with 0,1 mm/rev as feed.
- **Strategy 3.** The bar is turned along the entire length at a diameter D_A , then, the tool is repositioned and the length B is machined to a diameter D_B . Feed is equal to 0,2 mm/turn. The repositioning time is equal to 2s.



- **Strategy 4.** Same as Strategy 3, with feed 0,1 mm/turn.

It required to:

- a) Determine the feasibility of each strategy.
- b) For feasible strategies, determine the maximum bending.
- c) State which one, among the feasible strategies, has the highest throughput (assuming the same values for fixed times).

depends on FORCE, length outside chuck... ecc



different STRATEGY respect the length of WORKPIECE...



Example 6

13

Maximum RPM	6000 rpm
Efficiency	0,95
Kronenberg coefficient x	0,29
$k_{c0,4}$	2200 MPa
distance “c”	5 mm
Young’s module E	206000 MPa
Cutting speed	260 m/min
Entering angle of main cutting edge	90°
Overtravel	2 mm
Nominal lathe power	7,6 kW

HERE we are going to use a particular device of TURNING CENTER

common on Lathe $\Rightarrow$ **LOADER** to work portion of long metal bar:

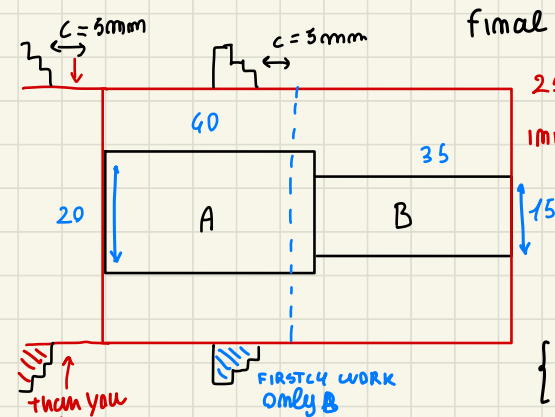
to feed the BAR you have an hole in the middle of the three

Jaw chuck to push the part in the working space...

open Jaws and reload the BAR $\rightarrow$ Automatic LOADER



We are in this particular condition



Initial BAR to machine

you push the long
BAR inside the envelop
of WORKING

$D_o = 25$ initial BAR Diameter

{ CYLINDER A: 25 $\rightarrow$ 20
CYLINDER B: 25 $\rightarrow$ 15

then you
WORK A

you have two STRATEGIES to perform this operation

(STRATEGY 1): push the BAR such to WORK only cylinder B

so three JAW-chuck enables only to WORK B

and after machined B (cutted) then the WORK HOLDING open and
push the BAR to WORK on different position

secondary
removal



initial removal

$\rightarrow$ external cylindrical turning

(STRATEGY 2)

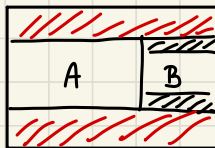
Re-positioning we fix the part and we can

WORK everything with different feed respect STR 1

(STRATEGY 3)

(STRATEGY 4)

$\uparrow$
with different
feed



firstly we cut all respect B
than we finish around B

$\Rightarrow$ so we are using an internal LOADER (bar loader) in two
ways $\rightarrow$ using full length (3,4) OR just by portion (1,2)

DATA: ST1: $f = 0.2 \text{ mm/REV}$ | ST3: $f = 0.2 \text{ mm/REV}$
 ST2: $f = 0.1 \text{ mm/REV}$ | ST4: $f = 0.1 \text{ mm/REV}$

$\Delta T_{\text{REPOSITIONING}} = 2 \text{ s}$ (open chuck, push, close chuck)

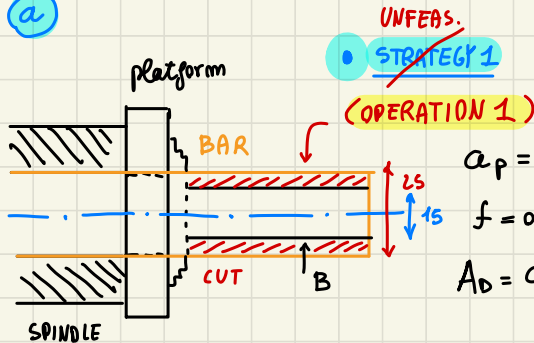
$n_{\text{MAX}} = 6000 \text{ rpm}$

$P_e = 7.6 \text{ kW}$ $\eta = 95\%$

$C = 5 \text{ mm}$ (both situations)

Let's discuss the different strategies

(a)



~~UNFEAS.~~

STRATEGY 1

(OPERATION 1)

$a_p = \frac{25 - 15}{2} = 5 \text{ mm}$ (relevant Roughing)

$f = 0.2 \text{ mm/REV}$

$A_D = a_p \cdot f = 5 \cdot 0.2 = 1 \text{ mm}^2$

$F_c = 2200 \left(\frac{0.4}{0.2 \sin(90^\circ)} \right)^{0.23} \cdot 1 = 2689.8 \text{ N}$

from F_c we evaluate:

$\Rightarrow P_c = F_c \cdot v_c = \frac{2689.8 \cdot 260}{60 \cdot 1000} = 11.7 \text{ kW}$
 [to have [kW]]

TO COMPARE with

$> 7.6 \cdot \eta = 7.6 \cdot 0.95 = 7.22 \text{ kW}$

UNFEASIBLE OP!

already OP1 is NOT

feasible for ST1 ... $\Rightarrow$ so neither A is evaluated OP2..

(STOP!)

(FEASIBLE) ✓

STRATEGY 2

all equal except for $f = 0.1 \text{ mm/REV}$

(OPERATION 1)

so changing

$A_D = 0.5 \text{ mm}^2$

$F_c = 1644.33 \text{ N}$

$\Rightarrow P_c = 7.125 \text{ kW} < 7.22 \text{ kW}$ OK FEASIBLE from POWER Requirement ✓

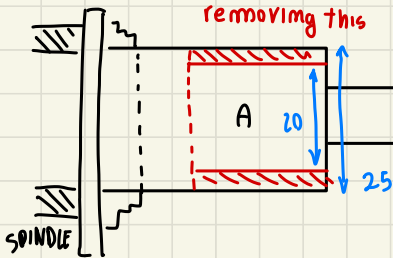
we need to check only $n < n_{\text{MAX}}$ for Feasibility

↓
ST2, OP1 feasible!
↑

$n = \frac{1000 \cdot 260}{\pi \cdot 15} = 5570 \text{ rpm} < 6000 \text{ rpm}$ OK feasible for Rpm

(OPERATION 2)

following situation
(already machined B)



$$f = 0,1 \text{ mm/REV}$$

$$a_p = \frac{25 - 20}{2} = 2,5 \text{ mm}$$

$$A_D = 2,5 \cdot 0,1 = 0,25 \text{ mm}^2$$

$$F_c = 2200 \left(\frac{0,4}{0,1 \cdot \sin(90)} \right)^{0,29} \cdot 0,25 = 822,17 \text{ N}$$

↓

$$P_c = F_c \cdot V_c = \frac{822,17 \cdot 260}{60 \cdot 1000} = 3,56 \text{ kW} < 7,22 \text{ kW} \quad \text{OK feasible}$$

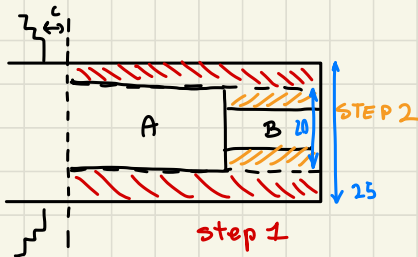
$$n = \frac{260 \cdot 1000}{\pi \cdot 20} = 4138 \text{ rpm} < 6000 \text{ rpm}$$

for shure on OP2, we have $D_{f2} < D_{f1}$ of OP1,
so $m_2 < m_1 < m_{MAX}$ FEASIBLE for shure!

OP2 also
FEASIBLE ✓

⇒ STRATEGY 2 OK, can be done all

• STRATEGY 3 (can 4 same str but different feed!)



(OPERATION 1)

$$f = 0,2 \text{ mm/REV}$$

$$a_p = 2,5 \text{ mm}$$

$$F_c = 22000 \left(\frac{0,4}{0,2 \cdot \sin(90)} \right)^{0,29} \cdot \underbrace{0,2 \cdot 2,5}_{A_D} = 1344,9 \text{ N}$$

↓

$$P_c = F_c \cdot V_c = \frac{1344,9 \cdot 260}{60 \cdot 1000} = 5,83 \text{ kW} < \gamma P_e \quad \text{OK}$$

$$n = \frac{1000 \cdot 260}{20 \pi} = n_{ST2, OP2} = 4138 \text{ rpm} < m_{MAX} \quad \parallel \text{OP1 feasible OK} \parallel$$

(OPERATION 2) here is easy to see how we are

cutting some amount of material
and use WORK in a similar
condition of already computed

$$P_c = 5,83 \text{ kW}$$

→ OK
↑

$$n = \frac{1000 \cdot 260}{15 \pi} = n_{OP1, ST1} < m_{MAX} = 5517 \text{ rpm}$$

|| STRATEGY 3
is fully feasible ✓

STRATEGY 4 (OPERATION 1)

$$f = 0,1 \text{ mm/REV} \quad a_p = 2,5 \text{ mm}$$

same computation of before... $P_c = 3,56 \text{ kW} < P_{e\eta}$ OK POWER

$$n = 4138 \text{ rpm} < n_{\max} \rightarrow \downarrow \text{OK, OP1 feasible}$$

(as ST3, OP1)

(OPERATION 2)

$P_c = 3,56 \text{ kW}$ cutting same material, equal req

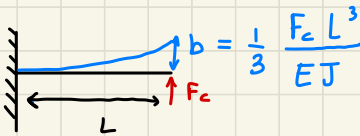
$$n = 5517 \text{ rpm} \quad \underline{\text{OK}}$$

|| STRATEGY 4 FEASIBLE! ||

SO ONLY ST1 IS NOT possible!
(because $P_c > \eta P_e$)

b) We have all DATA necessary
to compute BENDING

Three Jaw chuck fixturing:



$$b = \frac{1}{3} \frac{F_c L^3}{EJ}$$

$$J = \frac{\pi}{64} D^4$$

(initial diameter D , respect before cutting)

BENDING

$$b = \frac{64}{3} \frac{F_c L^3}{E \pi D^4}$$

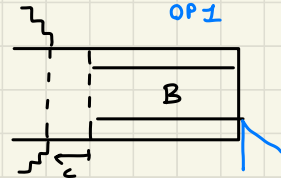
We have to consider
also C on the
full length for
bending.

STR2, OP1

$$b_{2,1} = \frac{64}{3} \cdot \frac{1644,33 (35+5)^3}{206000 \cdot \pi \cdot 25^4}$$

$$= 0,00888 \text{ mm} \approx 9 \mu\text{m}$$

$$\begin{cases} L_B = 35 \\ L_A = 40 \end{cases} \quad b_{2,1} \quad \text{OP1}$$

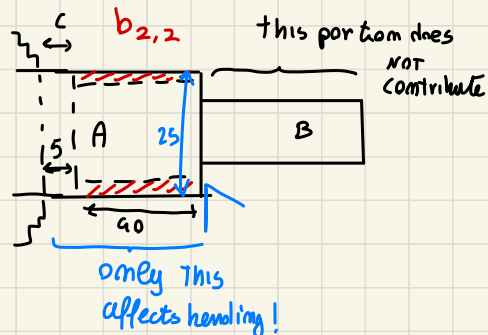


STR2, OP2

$$b_{2,2} = \frac{64}{3} \frac{822,17 \cdot (40+5)^3}{206000 \cdot \pi \cdot 25^4} =$$

$$= 0,00632 \text{ mm} \approx 6,32 \mu\text{m}$$

$b_{2,2} < b_{2,1}$
because $F_{c22} < F_{c21}$!
even if $L_{22} > L_{21}$



• **STR 3, OP1**

$$b_{3,1} = \frac{64}{3} \frac{1344,9 (34+40+5)^3}{206000 \cdot \pi \cdot 25^4} =$$

$$= 0,058 \text{ mm} = 58 \mu\text{m}$$

this times we work
with full length
 $L_B + L_A + C$

• **STR 3, OP2**

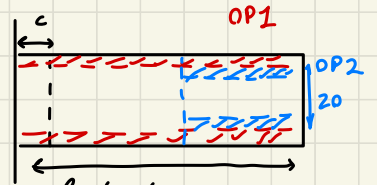
Relevant BENDING...
Bigger ERROR!

$$b_{3,2} = \frac{64}{3} \frac{1344,9 (34+40+5)^3}{206000 \cdot \pi \cdot 20^4} =$$

$$\Rightarrow = 0,14187 \text{ mm} = 142 \mu\text{m}$$

~ 0,3 mm error! Diameter! BAD

CUT from B to limit



$L = L_A + L_B + C$ always all
full length center cutting
starts from $D=20$

• **STR 4, OP1**

some FORCES acting of STR3, $\rightarrow$
the MOST CRITICAL is OP2 respect smaller D

$$\Rightarrow b_{4,2} = \frac{64}{3} \frac{822,17 (34+40+5)^3}{206 \cdot \pi \cdot 20^4} =$$

STR 4, OP2

$$= 0,0867 \text{ mm} = 87 \mu\text{m}$$



SUMMARIZE What happens during machine...

computing MACHING TIME:

considering in **STR 2** we have a repositioning time of $\Delta t = 2 \text{ s}$

$$t_{m,2} = \frac{35 + 5 + 2 + 2}{0,1 \cdot 5517} \cdot 60 + 2 + \frac{40 + 2,5 + 2 + 2}{0,1 \cdot 4138} \cdot 60 = 13,5 \text{ s}$$

↑
cutting portion B,
then exit moving radially of 5 mm to exit from workpiece (ap)
+ extra path IN/OUT (2 mm)

OP1: radial mot extra path
REPOSIT.
OP2
Full STR 2

STR 3

$$t_{m,3} = \frac{40 + 35 + 2,5 + 2 + 2}{0,2 \cdot 4138} \cdot 60 + 2 + \frac{35 + 2,5 + 2 + 2}{0,2 \cdot 5517} \cdot 60 = 10,2 \text{ s}$$

full length ap extra
JUST CUT B

STR 4

$$t_{m,4} = \frac{40 + 35 + 2,5 + 2 + 2}{0,1 \cdot 4138} \cdot 60 + 2 + \frac{35 + 2,5 + 2 + 2}{0,1 \cdot 5517} \cdot 60 = 18,3 \text{ s}$$

identical
to STR3 but
with different f

so:

$$\left\{ \begin{array}{ll} \bullet \text{ STR2 } t_{m2} = 13,5s & b_{\max2} = 9 \mu m \\ \bullet \text{ STR3 } t_{m3} = 10,2s & b_{\max3} = 142 \mu m \\ \bullet \text{ STR4 } t_{m4} = 18,3s & b_{\max4} = 84 \mu m \end{array} \right.$$

so (c) HIGHEST throughput guaranteed by the STR with lower t_m
 so STR3

BUT looking $b_{\max}$... we should decide STR 2 to
 avoid the big error due to handling higher om STR 3

↑
 the Requirement on accuracy affect the STR choice

↑
 OPERATION PLANNING on cylindrical turning with simple possible f , strategy
 ↓
 ON PROCESS PLANNING IT IS VERY COMPLEX TO FIND BETTER solution!